Demonstration of LLM-based In-situ Thought Exchanges for Critical Paper Reading

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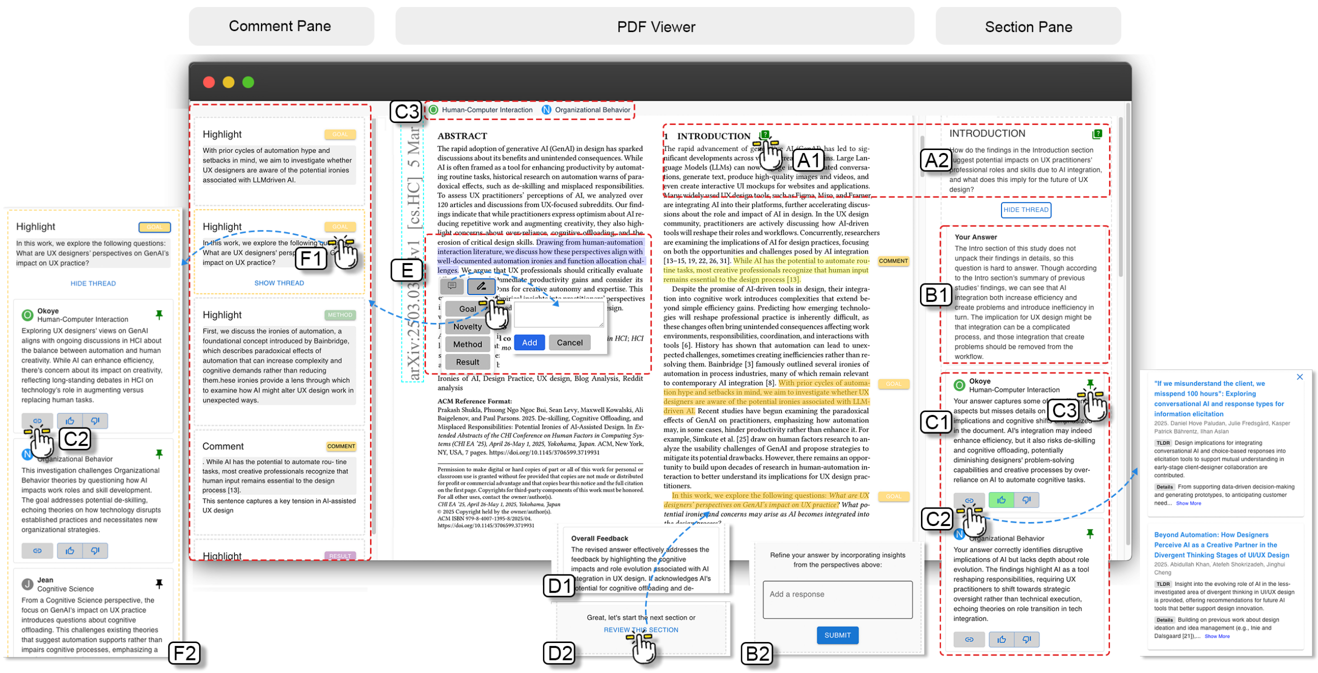

Figure 1: Our user interface contains three areas: the middle *PDF Viewer* to render PDF files; the left *Comment Pane*, to visualize highlighted text and comments; and the right *Section Pane*, which displays generated questions, participants' answers, and multi-agent feedback.

Abstract

Reading academic papers involves more than mere comprehension; it requires an active critical thinking process. However, most existing reading augmentation tools primarily focus on improving reading efficiency, may lead to cognitive offloading and over-reliance on AI systems. To explore how AI can be critically utilized to support academic paper reading, we design and demonstrate several features that foster critical thinking rather than promote cognitive offloading with the support of multiple AI agents.

Keywords

Critical Reading, Critical Thinking, Multi-Agent, Human-AI Interaction

1 Introduction

Reading academic papers is not merely about comprehending the content; it requires critical thinking abilities to engage in higher-order thinking skills such as analysis, synthesis, and evaluation [2]. This skill is especially important in interdisciplinary fields such as Human–Computer Interaction (HCI), which draws on knowledge from multiple disciplines, including computer science, social science, psychology, and design.

However, current LLM-based reading augmentation tools rarely explore how to foster readers' critical thinking abilities during the reading process. Instead, they tend to offload readers' cognitive effort by summarizing long texts such as TLDR (Too Long; Didn't Read), proactively locating key information [3], and generating direct answers to readers' questions. Increasingly, researchers are concerned that such tools may lead to over-reliance on AI and potentially degrade the critical thinking abilities of readers, especially junior students and early-career researchers [4].

Therefore, a key challenge of this work is how to leverage LLMs to design tools that actively support critical paper reading rather than replace the reader's cognitive engagement.

To address this challenge, we propose an LLM-based in-situ thought-exchange interface that supports researchers in critical reading while enabling them to exchange ideas with multiple agents offering diverse perspectives. In this paper, we present the design

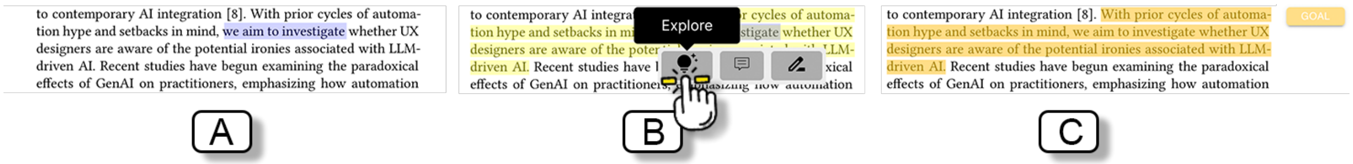

Figure 2: AI-highlighted text is not directly shown to readers. Once the reader selects the AI-highlighted text (A), an Explore button pops up (B), the reader can then check its semantic label (C) and proceed to further thought exchanges.

and key features of this interface. The source code, deployment instructions, and a live demo are available[1] .

2 Background

Academic papers are one of the most important mediums for research communication. With the widespread adoption of Large Language Model (LLM) tools, the number of scientific publications has increased year by year. Accordingly, an increasing number of LLM-based augmented reading tools have been proposed to improve readers' efficiency, for example, by quickly extract key information [3], or even delegating parts of the research process to AI systems. However, these developments raise growing ethical concerns. In particular, there is increasing worry that over-reliance on AI may hinder students' critical thinking abilities [4].

To address this challenge, several prior studies have explored enhancing readers' critical thinking through modern augmented reading interfaces. For example, Peng et al. designed CReBot [5] to interactively pose section-level critical thinking questions that guide junior researchers' reading processes. Similarly, Yuan et al. introduced CriTrainer [6] to support junior researchers in identifying a paper's main contributions and formulating critical questions. While these studies demonstrate that question-driven critical reading is an effective approach, such methods may still lead researchers into confirmation bias if critical reflection occurs without external perspectives or thought exchange.

To address this limitation, we designed an LLM-based in-situ Thought Exchange interface that facilitates structured interaction and perspective-sharing during the reading process.

3 Interface Design

3.1 Active Thought Exchanges

To encourage readers to think and reflect independently rather than over-relying on AI, we utilize the QRAC (Question, Read, Answer, and Check) approach [1] as the main interaction workflow in this work. Before reading each section, readers need to click the *Question* button (Figure 1-A1) beside the section title to generate critical thinking questions based on the current section context (Figure 1-A2). Then, readers need to keep the questions in mind while they *Read* the corresponding section and enter their initial *Answer* (Figure 1-B1). Once the reader finishes entering their response, the interface exchanges thoughts from three different agents' perspectives with the reader to *Check* (Figure 1-C1).

To further promote the reader's self-reflection ability while reading the paper, the interface asks the reader to integrate the agents' answers and improve their initial response (Figure 1-B2). Then, the reader receives overall feedback evaluating the improvement between their two answers (Figure 1-D1).

3.2 Interact with the Agents

To promote readers' self-reflection while reading the paper, we designed several interactive features that allow readers to interact with the agents. Readers can click the reference button (Figure 1-C2) to check relevant work that supports the agents' statements. They can also click the thumbs-up or thumbs-down button (Figure 1-C2, right) to agree or disagree with the agents' statements; the overall feedback will adjust based on the readers' reactions. Furthermore, if readers wish a specific agent to provide further input in the subsequent reading flow, they can click the Pin button (Figure 1-C3) to pin the agent to the top bar. The pinned agent will continue to appear in the following thought exchanges.

3.3 Active Reading Support

Following common active reading support features, we also provide comment and highlight functions. When the reader selects text, a toolbox pops up on the left side of the cursor (Figure 1-E). The reader can either add a comment to the selected text or highlight it. Once they choose to highlight, they can also attach a label from Goal, Novelty, Method, or Result. The label will be rendered in the margin of the page at the same height as the corresponding text. The highlighted text and comments are visualized as card components in the left Comment Pane.

3.4 Highlight Scaffolding

Nowadays, automatic highlighting is also widely used in commercial augmented reading interfaces to help readers locate the "key" sentences of a paper. However, it may lead to potential cognitive offloading in the readers' thinking process, which is particularly risky for junior students and researchers in developing their metacognitive skills.

To address this issue, we adopt automatic highlighting as scaffolding in the following two ways: (1) Once the user selects text that overlaps with AI-highlighted sentences, an Explore button will pop up, allowing the reader to check the semantic label of the highlight and proceed to further thought exchanges with the agents (Figure 2). (2) After users finish the QRAC workflow, a Review button will be shown in the Section Pane (Figure 1-D2). The reader can click this button to trigger the highlighting of key sentences in the current section and review whether they missed anything. All AI-highlighted sentences will also be added as information cards in the left Comment Pane.

[1] GitHub: https://github.com/Xinrui-Fang/InSituCPR ; Demo: https://insitucpr.org/

3.5 Passive Thought Exchanges

After the information cards are generated in the Comment Pane, the reader can further interact with them by clicking the label button in the upper-right corner of each card (Figure 1-F1). The agents will then reinterpret the highlighted text and comments from their disciplinary perspectives (Figure 1-F2).

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